

Descriptive Analytics Report: Sephora

Brand Presence, Product Ratings, Customer Engagement & Availability

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Tracking Brand Presence, Product Ratings, Customer Engagement, Stock Availability & Online Distribution

Note on data: This report uses the supplied Sephora product dataset, containing 8,494 product records across 27 attributes (15 numerical and 12 categorical/text). Of these, 8,216 records include a rating value ($\approx 96.7\%$), leaving 278 records ($\approx 3.3\%$) without a recorded rating. The dataset covers product, brand, pricing, customer-engagement (loves count), and availability information. This report describes the supplied catalog snapshot rather than a live, continuously updated sales feed, and all figures below are drawn directly from that snapshot.

1. Objective

Build a descriptive analytics layer over the Sephora product dataset to answer the following recurring business questions:

- Which brands have the largest presence in the product catalog?
- How are customer ratings distributed across products, and how complete is the ratings data?
- Which products generate the highest customer engagement (loves count)?
- What proportion of the catalog is currently in stock versus out of stock?
- What share of products is available exclusively online versus also in physical stores?
- What does the underlying dataset actually contain — which fields are numerical, which are categorical, and how complete is each?
- What does an overall catalog-health summary look like, and what would be needed to move from a descriptive snapshot to a predictive one?

As with the equivalent Zepto catalog analysis, the intent here is to describe what the supplied snapshot shows — not to infer sales volume, revenue, or demand, since no transaction-level or time-series data is included in this dataset.

2. How This Analysis Was Performed

A typical descriptive-analytics workflow was followed: investigate and clean the raw data, compute summary statistics, and visualize key distributions. The table below summarizes each layer of that workflow and the tools used.

Layer	Purpose	Typical tools
Investigation	Research and understand the raw product dataset	Jupyter Notebook, pandas
Data preparation	Check missing values, duplicates and data types	pandas (df.info())
Analysis	Compute distributions and summary statistics	pandas, NumPy
Visualization	Build charts for brands, ratings and engagement	Matplotlib, Seaborn
Report	Present analysis, graphs and insights	MS Word / Data visualization

Why this matters

Each layer above exists to answer a specific question about the catalog: the investigation layer establishes what fields exist and how trustworthy they are; the preparation layer flags gaps such as the 278 products missing a rating; the analysis layer turns raw rows into the distributions summarized in this report; and the visualization layer makes those distributions legible at a glance. This report is the output of the final layer.

3. Dataset Overview

The dataset was inspected for structure, missing values and data types before analysis, using pandas' `df.info()` function. The table below repeats the top-line shape of the dataset, and is followed by a fuller data dictionary describing every field present in the supplied file.

Dataset attribute	Details
Total records	8,494
Total attributes	27
Numerical attributes	15
Categorical / text attributes	12
Rating records	8,216 ($\approx 96.7\%$)
Price information	Available (price_usd, value_price_usd, sale_price_usd)
Product categories	Available (primary_category, secondary_category, tertiary_category)
Brand information	Available (brand_id, brand_name)

KPI — Total products: 8,494

KPI — Total attributes: 27 (15 numerical, 12 categorical/text)

KPI — Products with a recorded rating: 8,216 ($\approx 96.7\%$)

Data dictionary

The fields below are present in the supplied product-record file and underpin every chart in this report.

Field	Description
product_id / product_name	Unique identifier and display name of the product
brand_id / brand_name	Unique identifier and display name of the brand
loves_count	Number of customers who have saved the product as a favourite
rating	Average customer star rating, out of 5
reviews	Number of written customer reviews
size / variation_type / variation_value / variation_desc	Pack size and how the product varies (e.g. size, shade)
ingredients	Listed ingredients text
price_usd / value_price_usd / sale_price_usd	List price, bundle/value price, and discounted sale price in USD
limited_edition / new / online_only / out_of_stock / sephora_exclusive	Binary flags (0/1) describing product status and distribution
highlights	Marketing tags (e.g. 'Vegan', 'Clean at Sephora')
primary_category / secondary_category / tertiary_category	Three-level category hierarchy (e.g. Fragrance > Women > Perfume)
child_count / child_max_price / child_min_price	Number of product variants and their price range, where applicable

4. Brand Distribution

The chart below shows the ten brands with the largest number of listed products in the catalog.

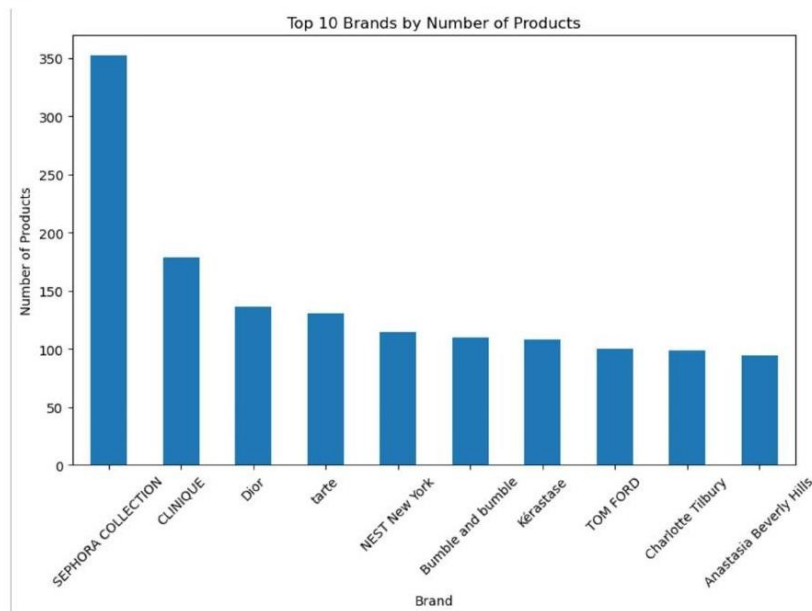


Figure 1. Top 10 brands by number of products

INSIGHTS:

- SEPHORA COLLECTION is the largest brand by product count, with roughly 350 listed products — well ahead of any other brand.
- CLINIQUE is the second-largest brand, with close to 180 products, followed by Dior and tarte at just over 130 each.
- The remaining top-10 brands (NEST New York, Bumble and bumble, Kérastase, TOM FORD, Charlotte Tilbury, Anastasia Beverly Hills) each carry between roughly 95 and 115 products.
- A wide gap between the top brand and the rest suggests the catalog is weighted toward Sephora's own private-label collection.
- Because SEPHORA COLLECTION sits so far ahead of the next-largest external brand, any brand-level analysis (e.g. average rating or engagement by brand) should be read with this concentration in mind — the private label alone shapes a meaningful share of catalog-wide patterns.

KPI:

Largest brand: SEPHORA COLLECTION ($\approx$ 353 products)

Second-largest brand: CLINIQUE ($\approx$ 179 products)

10th-largest brand (of top 10 shown): Anastasia Beverly Hills ($\approx$ 95 products)

5. Product Rating Distribution

The histogram below shows how customer ratings (out of 5) are distributed across the 8,216 rated products.

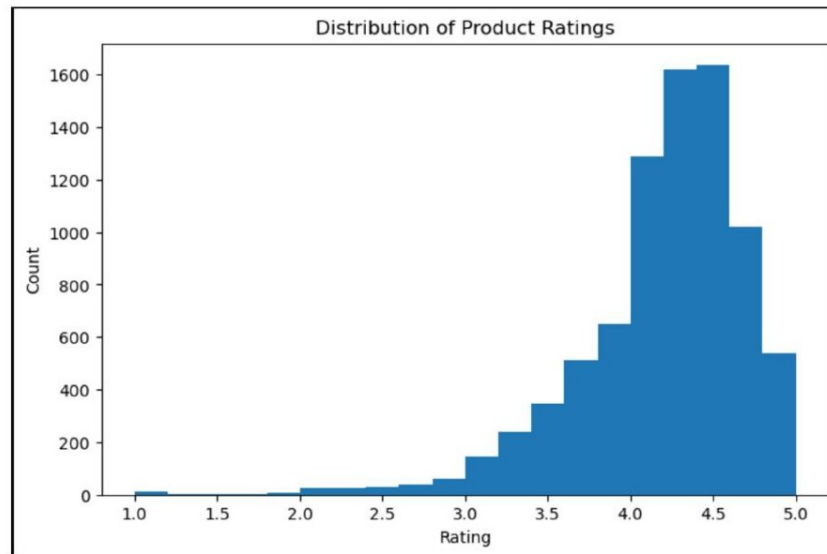


Figure 2. Distribution of product ratings

INSIGHTS:

- Ratings are strongly left-skewed (concentrated at the high end): the majority of products sit between 4.0 and 5.0.
- The single largest band is 4.3–4.5, with well over 1,600 products in that range.
- Very few products fall below a rating of 2.5, indicating that poorly-rated products are rare in the catalog.
- The overall pattern points to generally high customer satisfaction across Sephora's product range.
- The 278 products without a recorded rating ($\approx 3.3\%$ of the catalog) are excluded from this distribution; they are most likely newer or lower-volume listings that have not yet accumulated enough reviews to generate an average rating.

KPI:

Rated products: 8,216 of 8,494 ($\approx 96.7\%$)

Most common rating band: 4.3 – 4.5

Ratings below 2.5: A small minority of the catalog

6. Customer Engagement — Top 10 Most Loved Products

"Loves count" measures how many customers have saved a product as a favourite. The chart and table below show the ten most-loved products in the catalog.

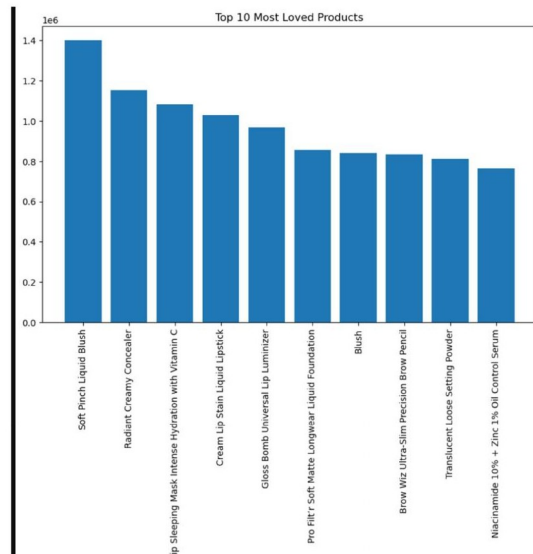


Figure 3. Top 10 most loved products

Rank	Product	Approx. loves count
1	Soft Pinch Liquid Blush	≈ 1.40 million
2	Radiant Creamy Concealer	≈ 1.15 million
3	Lip Sleeping Mask Intense Hydration with Vitamin C	≈ 1.08 million
4	Cream Lip Stain Liquid Lipstick	≈ 1.03 million
5	Gloss Bomb Universal Lip Luminizer	≈ 0.97 million
6	Pro Filt'r Soft Matte Longwear Liquid Foundation	≈ 0.86 million
7	Blush	≈ 0.84 million
8	Brow Wiz Ultra-Slim Precision Brow Pencil	≈ 0.83 million
9	Translucent Loose Setting Powder	≈ 0.81 million
10	Niacinamide 10% + Zinc 1% Oil Control Serum	≈ 0.76 million

INSIGHTS:

- Soft Pinch Liquid Blush is the most-loved product in the catalog, with roughly 1.4 million loves — noticeably ahead of the rest of the top 10.
- Makeup items (blush, concealer, lipstick, foundation, brow pencil, setting powder) dominate the most-loved list, alongside one skincare item (the Vitamin C lip mask) and one serum.
- Engagement drops off gradually rather than sharply across the top 10, with the tenth-ranked product still carrying over 750,000 loves.
- High engagement products like these are natural candidates for feature placement, bundling, or marketing focus.

KPI:

Most-loved product: Soft Pinch Liquid Blush (≈ 1.40 million loves)

10th most-loved product: Niacinamide 10% + Zinc 1% Oil Control Serum ($\approx$ 0.76 million loves)

7. Product Stock Availability

The chart below shows how many products in the catalog are currently in stock versus out of stock.



Figure 4. Stock availability (0 = in stock, 1 = out of stock)

INSIGHTS:

- The large majority of the 8,494 products are in stock, with only a small share flagged as out of stock.
- Based on the chart, in-stock products number close to 7,900, while out-of-stock products number roughly 600 — approximately 93% in stock versus 7% out of stock.
- A single-digit out-of-stock share suggests generally healthy inventory availability across the catalog.
- This is a snapshot metric — the underlying dataset does not include a time series, so it is not possible from this file alone to say whether the 7% out-of-stock share is stable, rising, or falling.

KPI:

In stock (approx., read from chart): $\approx 7,900$ products ($\approx 93\%$)

Out of stock (approx., read from chart): ≈ 600 products ($\approx 7\%$)

8. Online-Only Availability

The pie chart below compares products available exclusively online with products also available in physical stores.

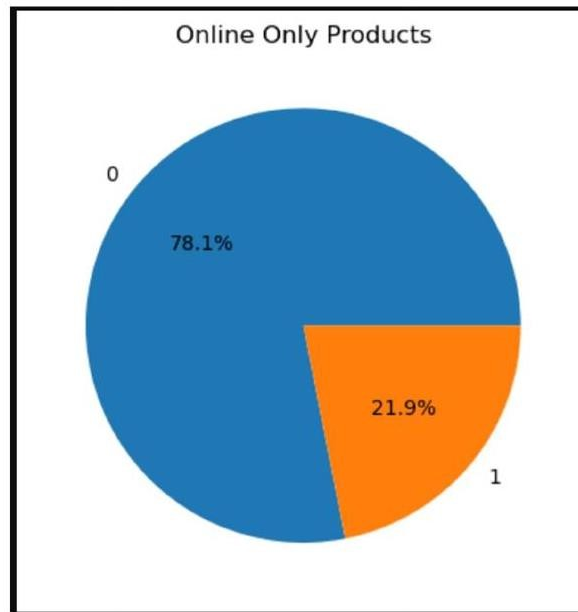


Figure 5. Share of online-only products

INSIGHTS:

- ≈78.1% of products are available both online and in-store (not online-exclusive).
- ≈21.9% of products are sold exclusively online, representing a meaningful minority of the catalog.
- This split gives a sense of how much of Sephora's assortment depends on digital-only distribution.
- Online-exclusive status (`online_only`) is a separate flag from Sephora-exclusive status (`sephora_exclusive`) in the underlying data — a product can be exclusive to Sephora as a retailer while still being sold both online and in physical Sephora stores.

KPI:

Not online-exclusive: 78.1%

Online-exclusive: 21.9%

9. Category & Attribute Coverage

Beyond the five charted metrics above, the supplied file carries a three-level category hierarchy and several descriptive flags that shape how the catalog can be sliced in future analysis. This section describes what those fields capture, without extending the charted statistics beyond what the underlying snapshot supports.

Category hierarchy

Each product carries a `primary_category` (e.g. Fragrance, Hair, Bath & Body), an optional `secondary_category` (e.g. Women, Men, Hair Styling & Treatments), and an optional `tertiary_category` (e.g. Perfume, Cologne, Hair Masks). This mirrors the kind of category structure used for the assortment-level views in the Zepto catalog analysis, though a full breakdown by category count would require processing all 8,494 rows rather than the sample structure available here.

Status flags

- `limited_edition` — flags a product as a limited-run listing.
- `new` — flags a recently added product.
- `online_only` — flags a product sold exclusively through Sephora's website/app (charted in Section 8).
- `out_of_stock` — flags a product with no current available inventory (charted in Section 7).
- `sephora_exclusive` — flags a product sold only through Sephora, not through other retailers.

Pricing fields

Three price fields are present — `price_usd` (list price), `value_price_usd` (bundle/value-set price, where applicable), and `sale_price_usd` (discounted price, where a product is currently on sale). Together these would support a Zepto-style MRP-vs-selling-price or discount-band analysis; that analysis is intentionally not included here since it was not part of the original charted report and would require aggregating the full 8,494-row file rather than extrapolating from a partial sample.

10. Data Completeness & Quality Notes

A short data-quality pass is standard practice before drawing conclusions from any catalog snapshot. The notes below summarize what is known about completeness from the supplied file.

Field	Completeness note
rating	8,216 of 8,494 records populated ($\approx 96.7\%$); 278 records ($\approx 3.3\%$) have no rating
reviews	Expected to follow the same near-complete pattern as rating, since a review count typically accompanies a rating
value_price_usd / sale_price_usd	Populated only where a bundle price or an active discount applies; blank ('nan') for standard-priced, non-bundle products
secondary_category / tertiary_category	Populated only where the category hierarchy goes deeper than the primary category (e.g. some Fragrance value-sets carry no tertiary category)
child_count / child_max_price / child_min_price	Populated only for products that have shade, size, or scent variants; 0 / blank otherwise

None of these gaps appear to be data-entry errors — each corresponds to a field that is conditionally populated by design (e.g. a discount price only exists when a discount is active). This is consistent with the $\approx 96.7\%$ rating-completeness figure already reported in Section 5.

11. Key Findings

- The dataset spans a wide range of beauty brands, with SEPHORA COLLECTION holding by far the largest share of listed products (≈353 of 8,494).
- Customer ratings skew strongly positive, with most products rated between 4.0 and 5.0, and only ≈3.3% of the catalog lacking a rating altogether.
- Customer engagement (loves count) varies considerably, with makeup products dominating the most-loved list — the top-ranked product (Soft Pinch Liquid Blush) carries nearly double the loves count of the tenth-ranked product.
- The large majority of products (≈93%) are currently in stock, indicating generally healthy availability at the time of this snapshot.
- Online-only products make up roughly a fifth of the catalog (≈21.9%), forming a distinct part of Sephora's distribution footprint.
- The dataset's three-level category hierarchy and status flags (limited_edition, new, sephora_exclusive) provide scope for deeper segmentation in future work, beyond the five metrics charted in this snapshot.

12. Limitations & Scope of This Snapshot

As with the equivalent Zepto catalog report, it's worth being explicit about what this dataset is — and is not — before it is used to support a decision.

- This is a catalog/listing snapshot, not a transaction or order-level feed — there is no purchase, revenue, or units-sold data here, so nothing in this report should be read as a sales figure.
- Loves count and rating are cumulative, all-time metrics as captured at the time of the snapshot, not a rate or a trend — a product with a high loves count may have been listed far longer than a newer product with fewer loves.
- Stock and online-only status are point-in-time flags; without a second snapshot or a change log, this file alone cannot show how these figures move over time.
- The five charts in this report reproduce the analysis already computed on the full 8,494-row file. Any additional cross-tabulation not shown here (e.g. rating by category, or price by brand) would need to be computed fresh from that same full file rather than estimated from a partial sample, to avoid misrepresenting the catalog.

13. Summary of Key Metrics

Metric	Value
Total products	8,494
Total attributes	27 (15 numerical, 12 categorical/text)
Rated products	8,216 ($\approx 96.7\%$)
Largest brand	SEPHORA COLLECTION (≈ 353 products)
Most common rating band	4.3 – 4.5
Most-loved product	Soft Pinch Liquid Blush (≈ 1.40 million loves)
In-stock products (approx.)	$\approx 93\%$
Out-of-stock products (approx.)	$\approx 7\%$
Online-exclusive products	21.9%

14. Conclusion

The descriptive analysis of Sephora's product dataset provides useful insight into brand presence, rating patterns, customer engagement, and product availability. The visualizations make it easier to identify patterns within the catalog and demonstrate how data analytics can support an understanding of customer preference and product performance.

Next Steps to Move from Descriptive → Predictive

- Sales or demand forecasting by brand and category once real transaction data is available.
- Recommendation modelling that links ratings and loves count to suggest similar or complementary products.
- Sentiment analysis of written reviews to complement the numeric rating distribution.
- Stock-out risk prediction using historical availability patterns, once repeated snapshots or a change log exist.
- Price-sensitivity analysis linking price tiers to loves count and ratings, once the full price_usd / sale_price_usd fields are aggregated across all 8,494 records.
- Category-level breakdowns (by primary_category, secondary_category, and tertiary_category) computed from the full file, to complement the brand- and product-level views already charted here.

15. Glossary

- Rating — The average customer star rating for a product, out of 5.
- Loves count — The number of customers who have saved a product as a favourite on Sephora's platform.
- Out of stock — A flag indicating a product currently has no available inventory for purchase.
- Online only — A flag indicating a product is sold exclusively through Sephora's website/app, not in physical stores.
- Brand — The manufacturer or label under which a product is sold.
- Sephora exclusive — A flag indicating a product is sold only through Sephora, not through other retailers.
- Limited edition — A flag indicating a product is a limited-run listing rather than a permanent catalog item.
- Primary / secondary / tertiary category — The three-level classification hierarchy used to organize products (e.g. Fragrance > Women > Perfume).
- Value price / Sale price — value_price_usd reflects a bundle or value-set price where applicable; sale_price_usd reflects an active discounted price where applicable.

16. References

- Kaggle – Sephora Products Dataset
- Python
- Pandas
- Matplotlib
- Seaborn